NAME: ____ CLASS: _________ INDEX: __________

CATHOLIC JUNIOR COLLEGE

JC2 PRELIMINARY EXAMINATION

Higher 2

MARKING SCHEME

BIOLOGY

9744/02

Paper 2 Structured Questions

24 August 2021

2 hours

Candidates answer on the Question Paper.

READ THESE INSTRUCTIONS FIRST

Write your name, class and index number in the spaces at the top of this page.

Write in dark blue or black pen.
You may use an HB pencil for any diagrams or graphs. Do not use staples, paper clips, glue or correction fluid.

Answer **all** questions in the spaces provided on the Question Paper.

The use of an approved scientific calculator is expected, where appropriate.
You may lose marks if you do not show your working or if you do not use appropriate units.

The number of marks is given in brackets [ ] at the end of each question or part question.

For Examiner's Use	
1 [15]	
2 [7]	
3 [10]	
4 [10]	
5 [13]	
6 [9]	
7 [7]	
8 [9]	
9 [10]	
10 [10]	
TOTAL	100

Answer **all** questions.

1 Fig. 1.1 shows an electron micrograph of a cell.

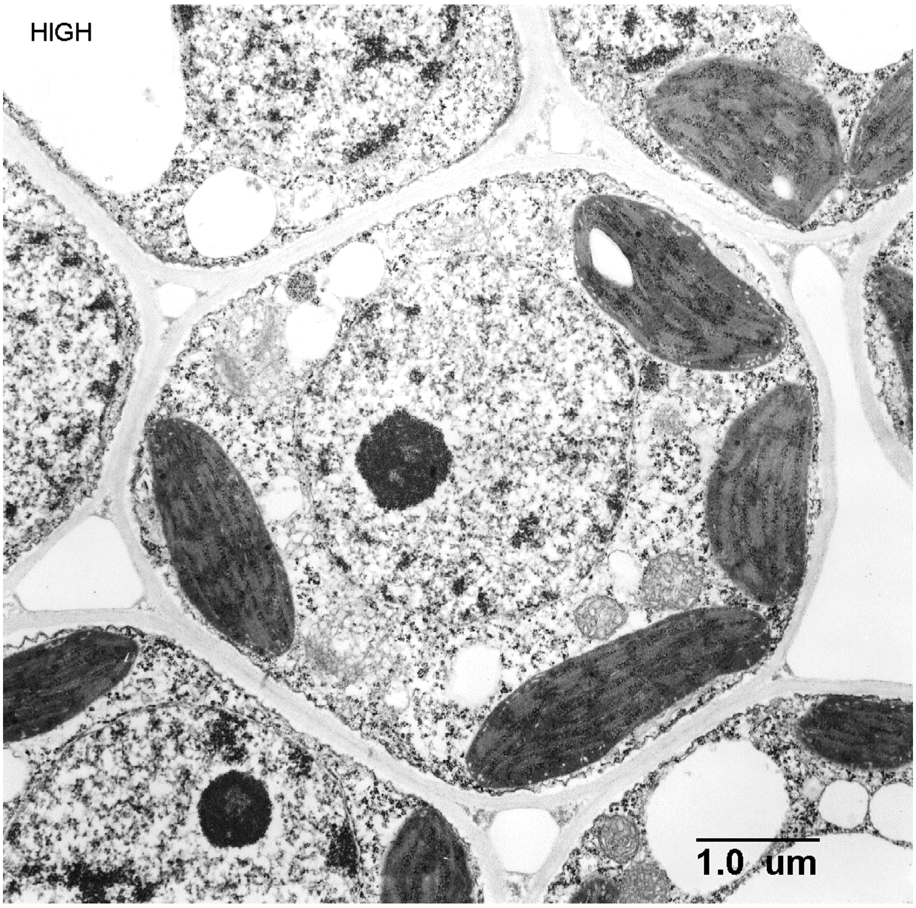


Fig. 1.1

(a) **(i)** With reference to the Fig 1.1, state two reasons why the cell is considered to be a eukaryotic cell.

Use 2 separate label lines, labelled **1** and **2**, to clearly mark out features visible in Fig. 1.1 to support your answer.

1 ……………………………………………………………………………………………

……………………………………………………………………………………………………

2 ……………………………………………………………………………………………

…………………………………………………………………………………………… [2]

1. Presence of a **true, membrane-bound nucleus** (that encloses the genetic material), unlike a prokaryotic cell where the genetic material lies freely in the cytoplasm (in a denser region called the nucleoid
2. Presence of **membrane-bound organelles** whereas these organelles are absent in a prokaryotic cell

Labels to point to the following:
1. Nucleus
2. Chloroplast or mitochondrion

[Reject: if label numbering and answer does not match.]

(b) Fig. 1.2 shows the structure of an adenovirus, which causes a wide range of illnesses such as common cold or flu-like symptoms, fever and sore throat.

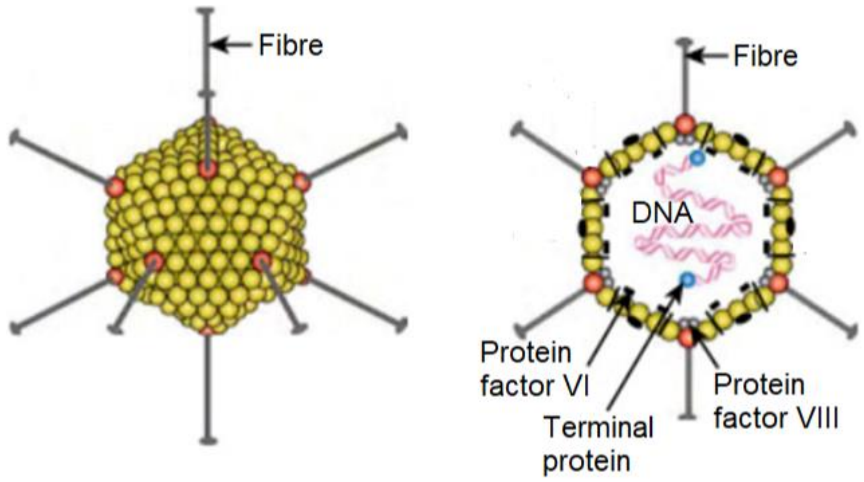


Fig. 1.2

(i) Outline the cell theory.

....

....

.... [1]

1. Cells are the smallest unit of life, all cells come from pre-existing cells, living organisms are composed of cells;;

(ii) With reference to Fig. 1.2, explain how this virus challenges the cell theory.

....

....

....

.... [2]

1. The virus may be considered living as it displays order and organised viral structures (capsid);;

Or

2. The virus may be considered living as it displays inheritance / hereditary characteristics due to presence of genetic material DNA;;

3. However, it is acellular / not a cell as there are no organelles;;

4. Viruses are not capable of replicating on their own without a host (must qualify)

Stachyose is a storage oligosaccharide found in beans, peas and other legumes.

Stachyose can only be metabolized by anaerobic microorganisms in the large intestine. It is responsible for the production of carbon dioxide, hydrogen sulfide and other gases which collectively make up what is known as flatulence.

The structure of stachyose is shown in Fig. 1.3.

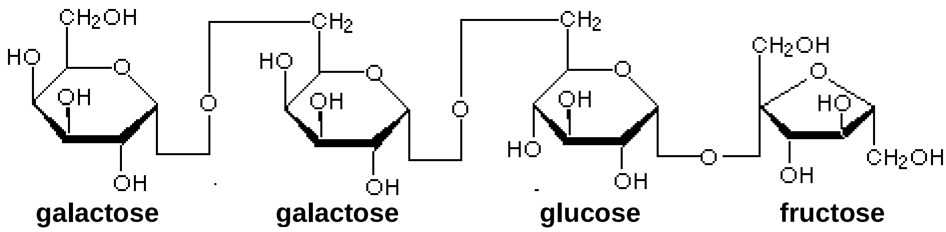


Fig. 1.3

(c) Define the term *glycosidic bond*.

....

.... [1]

1. A glycosidic bond is formed between **2 sugar residues / molecules** via a **condensation reaction** / removal of water;
2. when the anomeric **hydroxyl group of one sugar** unit react with another / any **hydroxyl group on another** sugar molecule

(d) Several commercial products are available to assist in the digestion of oligosaccharides like stachyose. Suggest how such products prevent the production of flatulence.

....

.... [1]

1. These products allow **the hydrolysis / cleavage / breakdown** of glycosidic bonds present in the oligosaccharides via **addition of a water molecule** before they become available to intestinal bacteria / anaerobic microorganisms;

(e) When small amounts of paper containing cellulose are ingested by children, there is usually no gas production. Suggest one reason and a possible consequence of such an event.

....

....

....

.... [2]

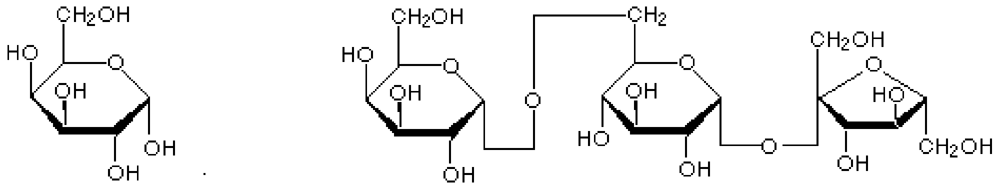
1. Cellulose is made up of **β-glucose** residues joined by **β-1,4 glycosidic bonds** which is not / cannot be broken down / digested in human intestines as humans lack the enzyme cellulase;
2. Intestinal blockage could occur OR The paper could pass out undigested or AVP

An enzyme known as α-galactosidase cleaves α-(1,6)-glycosidic bonds to sequentially release terminal galactose residues.

The breakdown of stachyose by α-galactosidase is a two-step process. In the first step of this process, stachyose is hydrolysed to galactose and raffinose.

(f) With reference to Fig. 1.3, draw the products of the first step in this hydrolysis.

[2]



Galactose **Raffinose**

1. Galactose correctly drawn and labeled
2. Raffinose correctly drawn and labeled

Fig. 1.4 illustrates the activity of α-galactosidase on the substrate stachyose over a period of time.

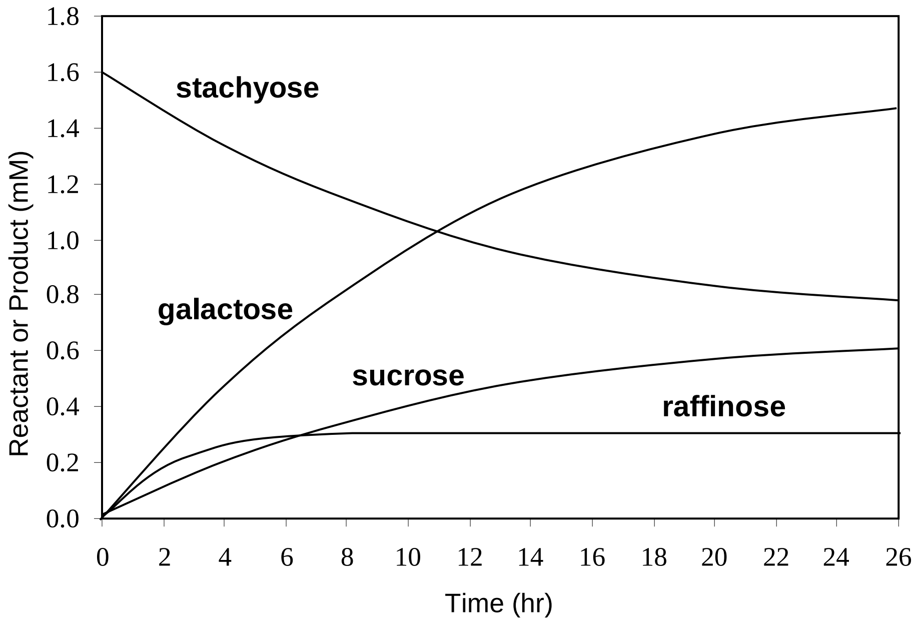


Fig. 1.4

(g) With reference to Fig. 1.4, explain why the level of raffinose remains relatively constant.

....

....

....

.... [2]

1. Stachyose is broken down into galactose and raffinose as shown by the drop in stachyose concentration from 1.6 mM to ~0.8 mM from 0 to 26 hours;
2. In the second step of hydrolysis, α-galactosidase is also able to hydrolyse / break down the α-1,6 glycosidic bond in raffinose, producing both galactose and sucrose as evidenced by the concentration of sucrose increasing from 0mM to ~0.6 mM from 0 to 26 hours;
3. At net equilibrium, both reactions produces a relatively constant raffinose concentration of 0.3 mM;

(h) Suggest why the rate at which stachyose is broken down decreases as the concentration of galactose increases.

……..

……..

……..

……………………………………………………………………………………………………… [2]

1. Galactose acts as a competitive inhibitor which is structurally similar to the stachyose substrate molecule and/or complementary in shape to the active site of the enzyme
2. It competes with stachyose for access / by binding to the active site; hence slowing down the rate of stachyose hydrolysis.

OR

3. Galactose acts as a non-competitive inhibitor / end product inhibitor where it binds to an allosteric site
and changes the three dimensional conformation of the active site of some α-galactosidase enzyme molecules
4. Such that there is a reduction of effective α-galactosidase / α-galactosidase can no longer bind to stachyose; hence slowing down the rate of stachyose hydrolysis.

[Total: 15]

2 **(a)** The "trombone" model of DNA replication postulates that two DNA polymerase enzymes work together in a protein complex.

In Fig. 2.1, structures labelled **P, Q** and **R** are enzymes involved in the process of DNA replication. Connecting proteins join the two DNA polymerase molecules to enzyme **P**, forming the protein complex.

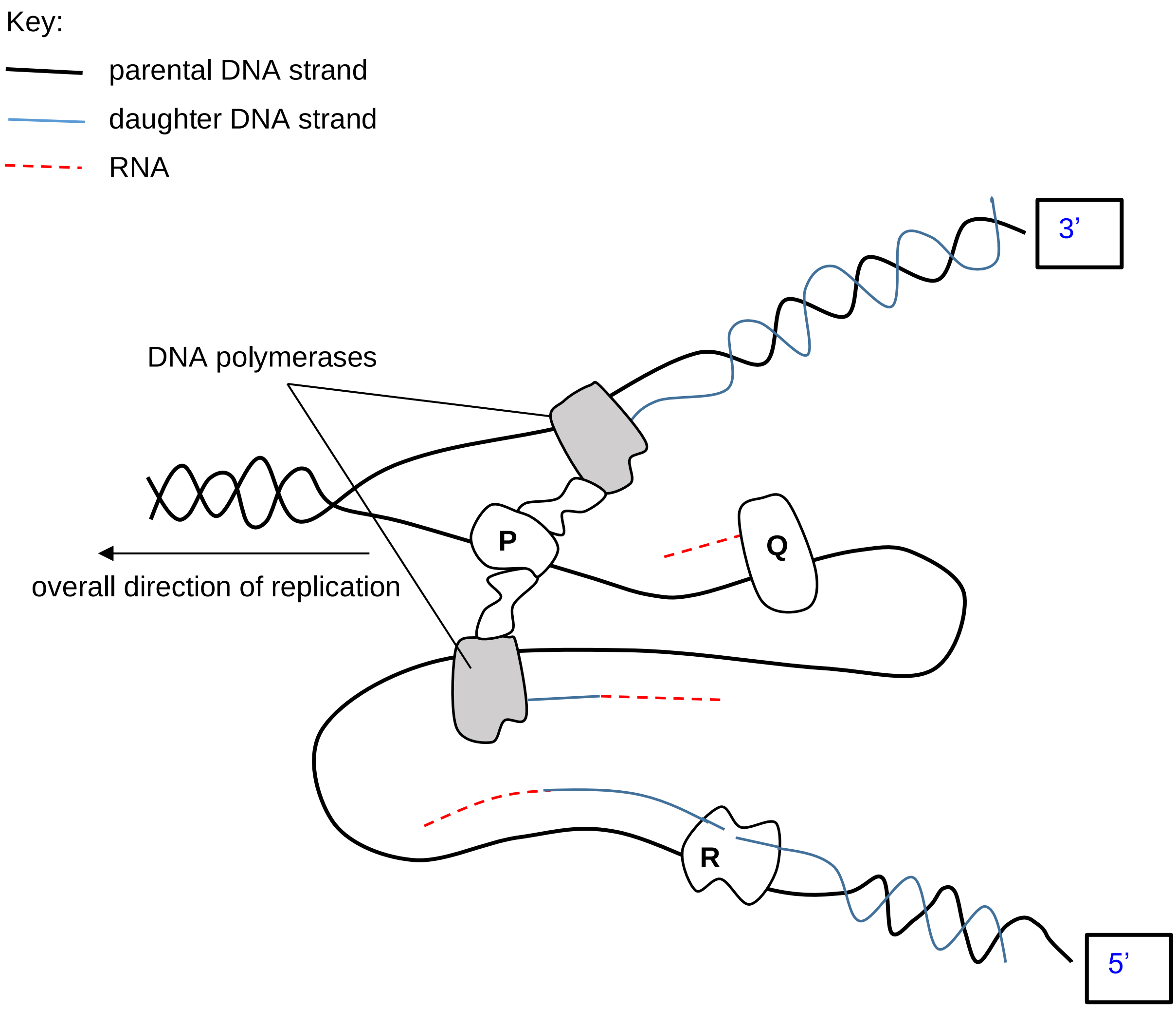


Fig. 2.1

(i) Complete the boxes to indicate clearly the 5' and 3' ends of the parental strands shown in Fig. 2.1. [1]

(ii) Identify enzymes **P, Q** and **R**. [1]

P: ……………………………………………………………………

Q: ……………………………………………………………………

R: ……………………………………………………………………

P: DNA Helicase

Q: Primase

R: DNA Ligase

Fig 2.2 illustrates the binding of a protein to a certain region of a DNA molecule

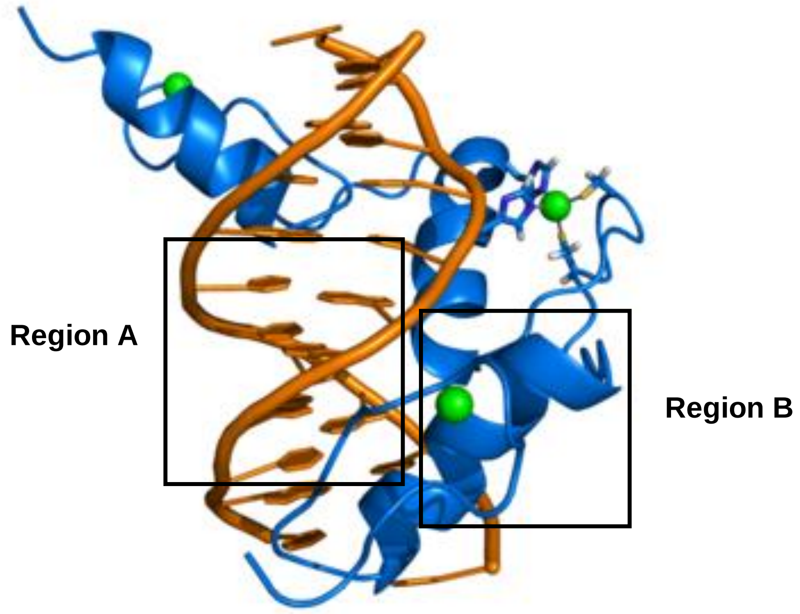


Fig 2.2

(b) With reference to Fig. 2.2, describe how the helical structure shown in **Region A** differs from that of **Region B**.

……

……

……

……

……

…… [3]

1. Helical structure (of the DNA) in region A is **double-stranded** unlike the **single-stranded** (polypeptide chain) in region B.
2. For Region A, each turn of the helix is made up of **10 base pair**; for Region B, the helix has **3.6 amino acids per turn**.
3. Helical structure in Region A is maintained by **phosphodiester linkages between deoxyribonucleotides** and **hydrogen bonds between complementary bases** whereas the helical structure of Region B by **peptide linkages** and **hydrogen bonds between amino acids**

   R: A is made up of deoxyribonucleic acid monomers while B is made up of amino acid monomers. Does not answer for why the structure is *helical*

Fig. 2.3 is a diagram of a telomere showing a unique single stranded G-rich 3’ overhang that loops back to form a hairpin structure.

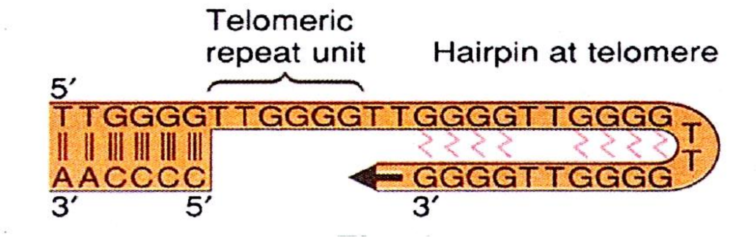


Fig.2.3

(c) Outline the role(s) of telomeres in DNA replication.

………

………

………

…………………………………………………………………………………………………… [2]

1. Telomeres play a protective function and serve to buffer the end replication problem (with every DNA replication) by delaying the degradation of genes near or closed to the end of the telomeres.
2. They provide a counting mechanism for cells to either stop dividing or to undergo apoptosis when the telomeres become too short.
3. AVP that is within the given context – in DNA replication.

[Reject: To prevent ends of chromosomes from fusing which makes a genome unstable – not within given context]

[Total: 7]

3 Fig. 3.1 shows the structure of molecules and some associated processes found in a eukaryotic cell.

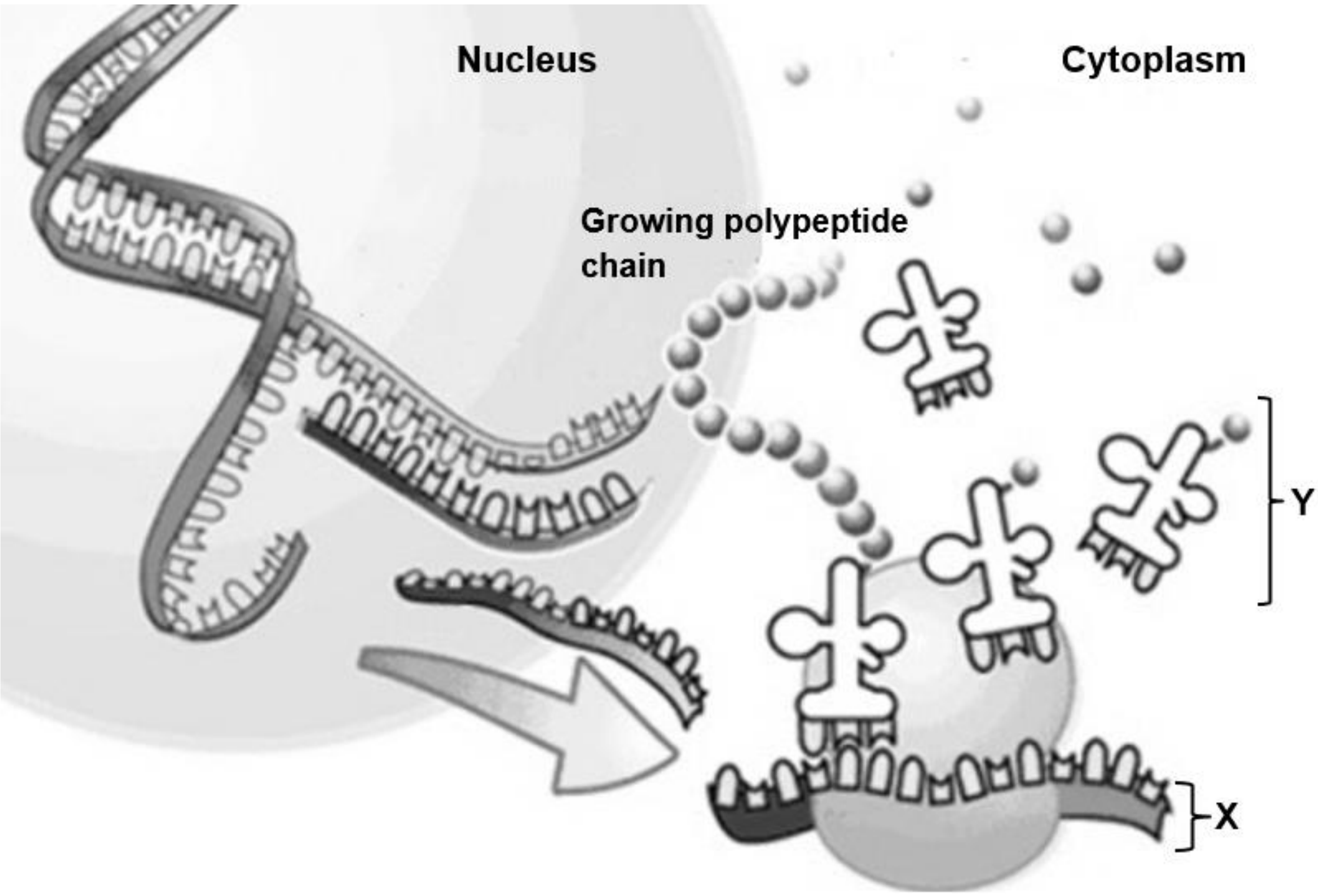


Fig. 3.1

(a) With reference to Fig. 3.1; outline the events that occurred in the nucleus that synthesised structure **X**.

....

....

....

....

....

....

....

.... [4]

1. **RNA polymerase II** binds to **promoter** of specific gene, forming a **transcription initiation complex** together with **general transcription factors**;
2. (a) RNA polymerase **reads the DNA template strand** from **3' to 5' direction** to form **pre-mRNA** / adds **ribonucleotides** to the **3'OH end** of the **pre-mRNA strand** or in **5' to 3' direction**;

OR

(b) RNA nucleotides / ribonucleotides are added via **complementary base pairing** with the **DNA template**;

OR

(c) **Phosphodiester bonds** are formed between ribonucleotides to form pre-mRNA;

3. (a) RNA **polymerase** continues with the transcription process until the **transcription stop site** is reached where the enzyme subsequently **dissociates** from **RNA-DNA**.

OR

(b) RNA polymerase II transcribes a **3'-TTATTT-5' sequence** (= the **polyadenylation signal**) **on the DNA** into **5'-AAUAAA-3'** along the **pre-mRNA**. At **a point about 10 to 35 nucleotides downstream from the AAUAAA signal**, proteins that are associated with the growing RNA transcript cut the RNA transcript free from the RNA polymerase. This **releases the pre-mRNA**.

4. **RNA processing** + **(any 1 of the 3):** **7-methylguanose** / **modified guanine cap** added to the **5' end** of the pre-mRNA strand / **3' poly(A) tail added to the 3' end of the pre-mRNA** / **RNA splicing** to synthesise **mRNA**.

(b) Identify structure **Y**.

.... [1]

1. **Aminoacyl-tRNA / charged tRNA** R: tRNA/ amino acids

(c) Parasites cause diverse types of disease, for example, malaria.

Drug treatments are required to address varying causes of pathogenesis. A new class of drugs has been deployed to block the formation of structure **Y** in parasites.

Suggest how the use of such drugs is effective in treating parasites infections.

....

.... [1]

1. Specific amino acids cannot be brought to the ribosomes, protein synthesis cannot occur causing death of parasites.

There is an absent of nucleus in bacteria. In addition, genes that are involved in the same metabolic pathway are usually arranged together in a structure called operon.

An example of an operon is the *lac* operon.

(d) Explain how the location of the *lac* operator allow it to regulate the expression of structural genes in *lac* operon.

……..

……. [1]

1. The operator is **located in between the promoter and structural genes / downstream the promoter and upstream the structural genes** (OWTTE) in *lac* operon;
   Either:
   this **prevents RNA polymerase** from **accessing / transcribing the structural genes** when **active lac repressor binds** to the operator.
   OR:
   this **allows RNA polymerase** to **access / transcribe the structural genes** when **no active lac repressor is bound** to the operator.

(e) Explain how the presence of high concentration of glucose results in low expression of structural genes found in *lac* operon despite the absence of any active *lac* repressor proteins.

……..

……..

……..

……. [2]

1. High concentration of glucose leads to **low concentration of cAMP**; **few cAMP binds to CAP protein**;
2. **Low concentration** of **cAMP-CAP complex / active CAP** that binds to **CAP-binding site**; therefore **RNA polymerase binds with low affinity to the *lac* promoter**.

(f) Some bacteria are found to have a loss of function mutation in the *lac* operator.

When grown in an environment containing high concentration of glucose and no lactose, these bacteria grow slower than the bacteria carrying normal functioning *lac* operator. Suggest why.

……..

……. [1]

ANY 1 of the following:

1. The mutation results in **constitutive / continuous expression of structural genes** in *lac* operon despite the absence of lactose, **wastage energy / resources**.
2. Energy / resources are channelled for the **synthesis of enzymes involved in lactose metabolism**. Therefore, **less energy / resources are channelled for growth**.
3. AVP

[Total: 10]

4 Epidermal growth factor (EGF) is a protein composed of 53 amino acids. It binds to specific cell surface membrane receptors to promote cell proliferation.

(a) Based on the information above**,** explain why EGF cannot move across cell surface membrane through channel proteins.

……

…………………………………………………………………………………………………...... [1]

1. EGF is composed of **53 amino acids** which is **too large** in size to pass through the **pore/hydrophilic channel** of any channel proteins.

HER2 receptors are glycoproteins found on cell surface membrane. They are a type of tyrosine kinase receptor that recognises EGF.

(b) Explain how the structure of HER2 receptor allows it to be embedded across cell surface membrane and serve as a transmembrane protein.

……..

……..

……..

……..

……..

……..

……..

……. [4]

1. The **transmembrane region** of **HER2 receptor** composed of **amino acids with non-polar / hydrophobic R-group**;
2. allowing the transmembrane region of HER2 receptor to interact via **hydrophobic interaction** with the **fatty acids / hydrocarbon tail** of **phospholipid** molecules.
3. The **regions of HER2 receptor** that are **facing** the **aqueous intracellular and extracellular environment** composed of **amino acids with either polar or charged / hydrophilic R-group**;
4. allowing these regions of HER2 receptor to interact via **hydrophilic interaction** with the **environment** / with the **phosphate heads** of the **phospholipid** molecules.

(c) Name another ligand molecule that binds to tyrosine kinase receptors and regulate blood glucose level.

……. [1]

1. Insulin.

Fig. 4.1 below briefly illustrates HER2 signalling pathway.

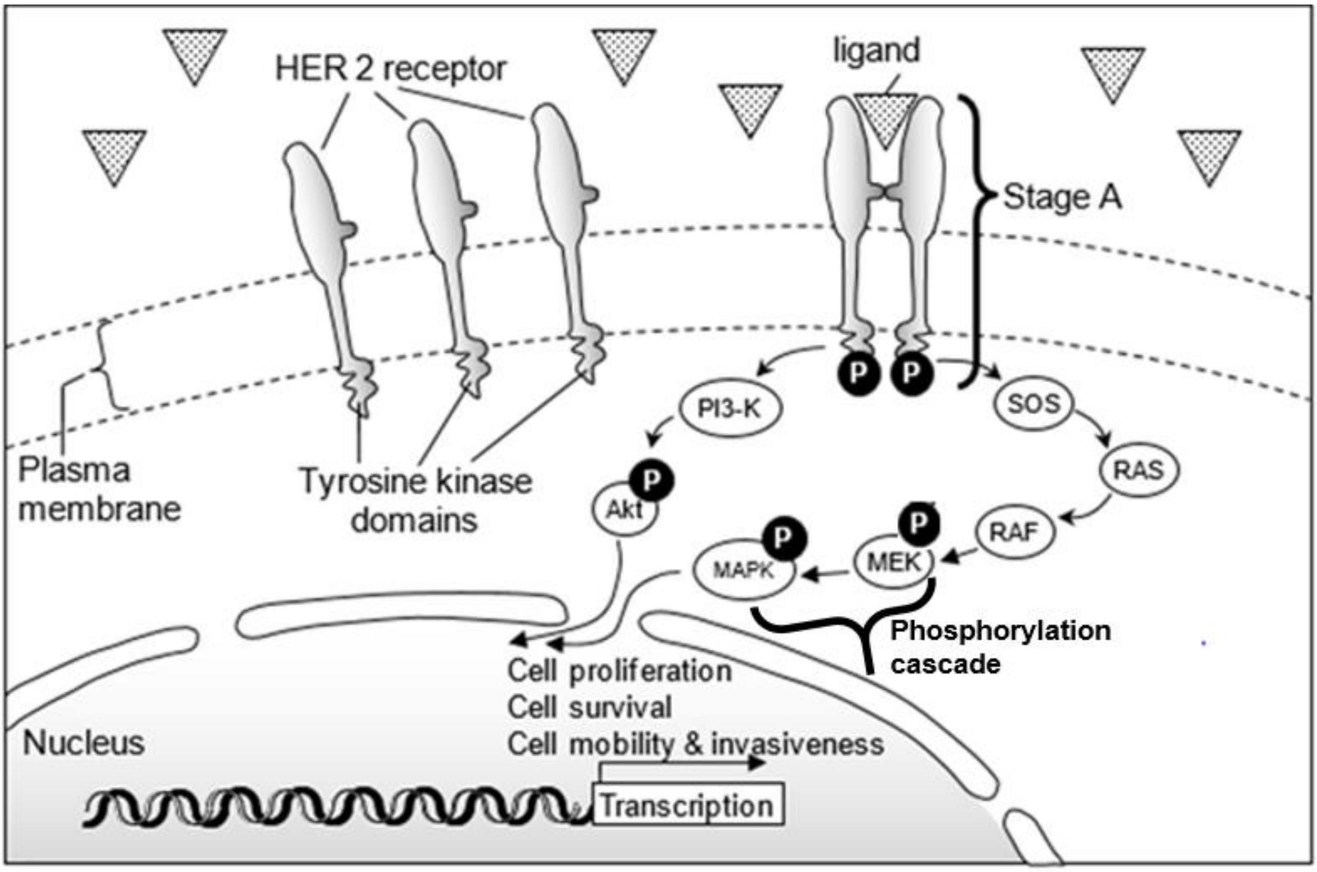


Fig. 4.1

(d) Explain the events occurring in stage A of the HER2 signalling pathway shown in Fig. 4.1.

…………………………………………………………………………………………………….

…………………………………………………………………………………………………….

…………………………………………………………………………………………………….

…………………………………………………………………………………………………….

…………………………………………………………………………………………………….

……………………………………………………………………………………………… [3]

1. **Ligand / EGF bind**s to **two HER2 receptor monomers**; as the **ligand / EGF and the ligand binding site of HER2 receptor** have a **complementary shape / structure**.
2. HER2 receptors dimerise; leading to the activation of the tyrosine kinase domain on each monomer.
3. Activated tyrosine kinase domain on each monomer catalyses **auto-phosphorylation** of **tyrosine** residue on the other monomer; leading to **activated HER2 receptor**.

(e) RAF (an example of protein kinase) can be activated by phosphorylation. It can be activated by direct interaction with its allosteric site too.

Fig. 4.1 shows that RAS protein activates RAF by direct interaction instead of phosphorylation.

Suggest why RAS does **not** activate RAF by phosphorylation.

………..

……… [1]

1. RAS protein is **not** a protein **kinase**; it is **unable** to catalyse phosphorylation.

[Teachers' Note: students have encountered Ras protein in Topic on Cancer and they should know that Ras protein is a type of G-protein]

[Total: 10]

5 Fig. 5.1 shows the role of stem cells in the immune system and the variety of cell types they are able to differentiate into. Fig 5.2 shows a functional group found in the nucleus of eukaryotic cells.

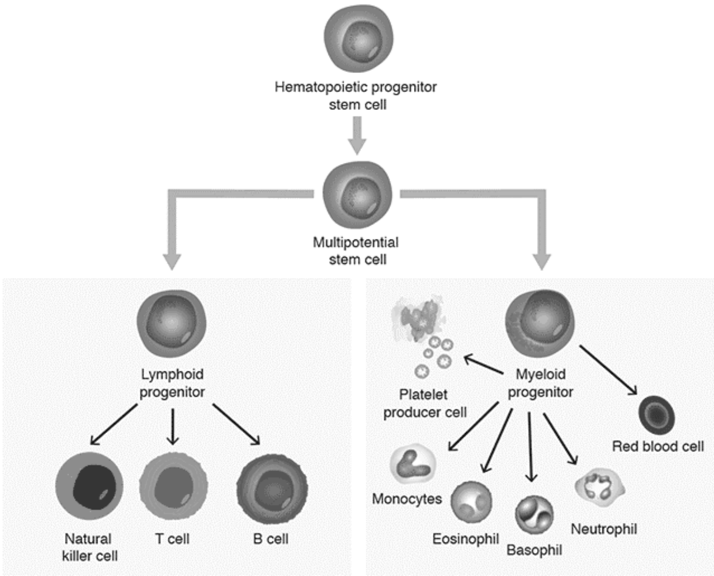


Fig. 5.1

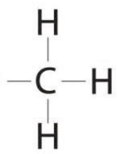


Fig. 5.2

(a) Name one source of totipotent stem cells.

....

....[1]

1. Cells from the initial divisions by the zygote / Morula / Blastomeres. OWTTE

(b) Explain the role of the functional group shown in Fig. 5.2 in determining the differentiation of stem cells into specific cell types.

....

....

....

....

....

....

....

.... [4]

1. Fig 5.2 shows a Methyl group which is used in **DNA methylation**.
   Followed by
2. Except for genes **specific to the specialised cell** and **house keeping** genes, all other genes in the genome are silenced during differentiation to specialised cell.
3. Each **cytosine** in **DNA** on both strands have a **methyl group added**.
4. Resulting in the inhibition of the action of RNA Polymerase, terminating transcription.
5. Results in Histone compacting and further silencing of gene groups. OWTTE

Max 3

or

1. genes **specific to the specialised cell** and **house keeping** genes, all other genes in the genome are silenced during differentiation to specialised cell.
2. Each **cytosine** in **DNA** on both strands have a **methyl group removed / DNA demethylation**.
3. Resulting in the removal of methyl group allowing the action of RNA Polymerase, allowing transcription.
4. Demethylation also results in Histones being less compacted a step closer to linear DNA ready for transcription. OWTTE

Max 3

(c) With reference to the control of cell cycle, explain how the initiation of cell division is controlled in stem cells.

……

……

……

……[2]

1. Initiation of cell division is controlled by the **G1 Checkpoint**
2. **Cell division is controlled** by the presence of the **Human Growth Hormone**
3. And is inhibited by contact inhibition. (less significant)

Max 2

(d) In an active organ such as liver, the rate of transcription in the cells is usually high.

With reference to the control of gene expression at transcriptional level, explain fully how the high rate of transcription in these cells can be achieved.

……

……

……

……

……

……

……

……[4]

1. **Distal control elements** in the form of **enhancers** which are **DNA sequences** upstream to the promoter.
2. **Activator proteins bind** complimentarily **to the enhancer sequences** (Enhancer/activator complex)
3. **DNA bending proteins are recruited** (OWTTE) which allow both the enhancer / activator complex and promoter to come into close proximity and subsequently bind.
4. The final complex along with **RNA polymerase gives rise to a higher rate of transcription** of the gene.

(e) Suggest why specific transcription factors are considered an efficient way of control of gene expression in eukaryotic genes.

………

………

………

……[2]

1. **Activator proteins** are an **example of STF**, involved with the upregulation of transcription rate. OWTTE / Accept description for **repressor proteins** too.
2. These activator proteins are themselves governed by gene expression which is dependent on external factors affected by the environment.
3. In this manner STFs help to preserve the conservation of energy within the cell. OWTTE.

[Total: 13]

6 In human, hereditary breast and ovarian cancer syndrome (HBOC) is an inherited genetic condition that follows an autosomal dominant inheritance.

.

HBOC is associated with loss of function mutations in *BRCA* genes which are a group of tumour suppressor genes. The products of these genes are involved in repairing of damaged DNA and initiating apoptosis in cells where the DNA damage is beyond repair.

Fig. 6.1 below shows the percentage of occurrence of breast cancer and ovarian cancer in women who are *BRCA* mutation carriers (women with HBOC) and women in the general population (women without HBOC).

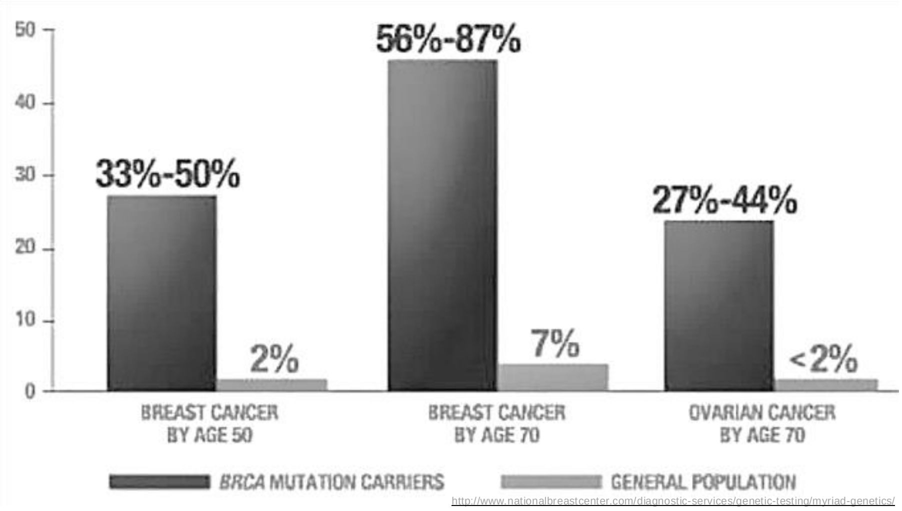


Fig. 6.1

(a) With reference to Fig. 6.1, explain the difference in percentage occurrence of breast cancer between women who are *BRCA* mutation carriers and women in the general population by the age of 50.

....

....

....

....

....

....

....

.... [4]

1. Women who are *BRCA* mutation carriers had a **higher percentage** of occurrence of breast cancer by **31% - 48% higher** as compared to women in the general population.
2. *BRCA* genes are **tumor suppressor genes**. *BRCA* mutation carriers having inherited mutated *BRCA* genes will be at an **increased risk of developing breast cancer**.
3. **Loss of function of both alleles** of *BRCA* genes leads to **non-functional BRCA proteins**.

Max: 1

4. **BRCA proteins are involved in DNA repair,** and with the loss of function there is **an increase in cells with DNA mutated.**
5. **Accumulation of mutations** in these cells may result in a Gain of function mutations e.g. proto-oncogenes to oncogenes OWTTE.

(b) With reference to the women in general population in Fig. 6.1, explain the difference in percentage occurrence of breast cancer between women by the age of 50 and by the age of 70.

……

……

……

……[2]

1. In a span of 20 years, there was seen a **5% increase in cancer**, this could be due to the **accumulation of mutations over time**. OWTTE
2. This may contribute to the multi-step of cancer progression due to loss of function mutation in tumour suppressor genes, gain in function mutation of proto-oncogenes, gain in function of telomerase gene, etc. OWTTE

(c) With reference to Fig. 6.1,

(i) describe the difference in percentage occurrence of ovarian cancer and percentage occurrence of breast cancer in women who are *BRCA* mutation carriers.

……

……………………………………………………………………………………………………….. [1]

1. There was a **lower percentage occurrence** in **ovarian cancer; 27% - 44%** as compared to the **percentage occurrence** in **breast cancer**, **56% - 87%** in women who are *BRCA* mutation carriers.

(ii) suggest a possible reason for the difference described in part **c(i)**

……

……………………………………………………………………………………………………….. [1]

1. The probability of developing ovarian cancer is naturally lower than the probability of developing breast cancer which can be **inferred from comparing the percentage occurrence of breast cancer and ovarian cancer in women in the general population** by the age of 70; 7% and less than 2% respectively. OWTTE

(d) Scientists concluded that women who are BRCA mutation carriers (women with HBOC) may not necessarily get ovarian cancer.

Use the information in Fig.6.1 to justify this conclusion.

………

……… [1]

1. By the age of 70, only **27% – 44% of women** who are *BRCA* mutation carriers (women with HBOC) suffer from ovarian cancer,

   OR:

2. By the age of 70, 56% - 73% of women who are *BRCA* mutation carriers (women with HBOC) does not suffer from ovarian cancer.

[Total: 9]

7 The figure below shows eukaryotic chromosomes.

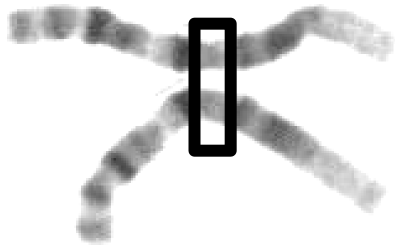


Fig 7.1

(a) Name the region shown and explain its role in full.

………

………

………

……….[2]

[L1] Source: Novel

1. **Centromere**
2. Cite of **spindle fiber attachment** at the **kinetochore**, involved in the movement of homologous chromosomes prophase / metaphase / anaphase OWTTE
3. Centromeres divide during anaphase to **facilitate separation of chromatids**.

Max 2

The Canine Genome Sequencing Project was successfully completed in year 2005. During the project, researchers needed to investigate two particular genes in dogs. The genes were the amino-methyl-transferase (*AMT*) gene (which codes for AMT enzyme to degrade excess glycine), and the T-cell leukemia translocation-associated (*TCTA*) gene (which codes for TCTA protein involved in osteoclastogenesis, the formation of osteoclasts for the breakdown of bone tissue).

(b) To increase sample size for statistical reliability, scientists selected several heterozygous female poodles for normal AMT phenotype and normal TCTA phenotype, and conducted test crosses on them. The following results were observed in the offspring.

Normal AMT, normal TCTA dogs	96
Normal AMT, mutant TCTA dogs	18
Mutant AMT, normal TCTA dogs	20
Mutant AMT, mutant TCTA dogs	94

(i) Using **A** and **a** to represent the alleles for the *AMT* gene, and **T** and **t** to represent the alleles for the *TCTA* gene, illustrate the above cross with the use of a genetic diagram.

[L2] Source: Modified ACJC 2013 Prelim/P2/Q4 [4]

Parental phenotype: normal AMT, normal TCTA x mutant AMT, mutant TCTA

Parental genotype: (AT) (at) x (at) (at)

1. P genotype, with brackets indicating linked genes;

or

Gametes (AT) (at) (At) (aT) (at)

2. 1m for circled gametes;

F1 genotype:

(AT) (at) (at) (at) (At) (at) (aT) (at)

F1 phenotype:

3. 1m for F1 genotype;

Normal AMT, normal TCTA Mutant AMT, mutant TCTA Normal AMT, mutant TCTA Mutant AMT, normal TCTA

4. 1m for corresponding phenotype;

REJECT: 1:1:1:1 phenotypic ratio

(ii) Explain the low frequency of recombinants observed in the offspring.

……………………………………………………………………………………………………

…………………………………………………………………………………………...……..[1]

1. Low frequency observed in recombinants is a reflection of the low occurrences of **crossing over** occurring between **non-sister chromatids of homologous chromosomes** during Prophase I.

[Total: 7]

8 Human T- lymphotropic virus type 1 (HTLV-1) is a retrovirus that brings about a form of cancer called T-cell leukaemia.

HTLV-1 has a similar reproductive cycle as human immunodeficiency virus (HIV) and predominantly infects CD4$^+$ T cells too.

Fig. 8.1 shows the structure of a mature HTLV-1 particle.

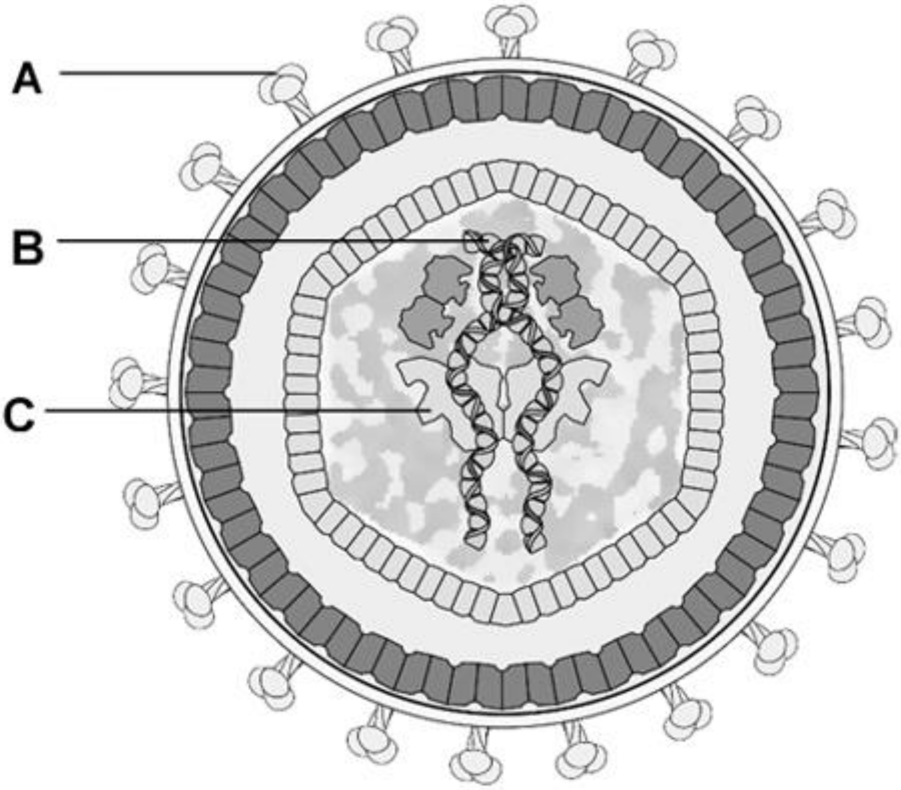


Fig. 8.1

(a) With reference to Fig. 8.1;

(i) Name structure **C**, which is the first viral enzyme involved in the reproductive cycle of HTLV-1

.... [1]

1. Reverse transcriptase

(ii) Describe the roles of structures **A** and **B** in the reproductive cycle of HTLV-1 in CD4$^+$ T cells.

....

....

....

.... [2]

1. Structure **A** binds to **<u>CD4⁺ receptor</u>** to facilitate **fusion** of membranes for the release of viral contents into host T cell;
2. Structure **B** serves as a **template** to form **<u>viral DNA</u>** for **integration** into **host** T cell **genome**.

(b) Suggest how HTLV-1 infection could lead to the onset of T-cell leukaemia.

……………………………………………………………………………………………………

…………………………………………………………………………………………………… [1]

1. Ref. to insertional mutagenesis.

(c) Explain why antibiotics are not used to treat HTLV-1 infection.

……………………………………………………………………………………………………

…………………………………………………………………………………………………… [1]

1. **Antibiotics** work against **bacteria** whereas **HTLV-1 infection** is caused by **virus**.

(d) Describe how HTLV-1 infection induces the production of antibodies in the body.

……………………………………………………………………………………………………

……………………………………………………………………………………………………

……………………………………………………………………………………………………

……………………………………………………………………………………………………

……………………………………………………………………………………………………

…………………………………………………………………………………………………… [3]

1. **Antigen presenting cells** such as macrophage and dendritic cells phagocytosed HTLV-1 / B cells take in HTLV-1 via receptor-mediated endocytosis (**<u>at least 1 example</u>**) and present the **epitope** on **MHC II molecule**.
2. **CD4+ T cells** that have **receptors specific / complementary to the epitope** presented on **MHC II molecule** will be activated into **Helper T cells**.
3. **Helper T cells** will **activate B cells** that **recognises the epitope**; activating it into **plasma cells** that produce **antibodies** that recognise the epitope.

(e) During the secondary immune response, majority of the antibodies secreted and circulating in the body is IgM.

Lumens of organs are mainly lined with epithelium. During secondary infection, majority of antibodies found within the lumen of organs are IgA instead of IgM. Suggest why.

……………………………………………………………………………………………………

…………………………………………………………………………………………………… [1]

1. IgA is able to be transported across epithelium but IgM cannot.

[Total: 9]

9 The polar bear (*Ursus maritimus*) is most closely related to the brown bear (*Ursus arctos*). The brown bear is widely distributed across Holoarctic habitats (land masses surrounding but not part of the Arctic Circle). The two species differ in morphology, ecology and behaviour, reflecting adaptations to different ecological niches.

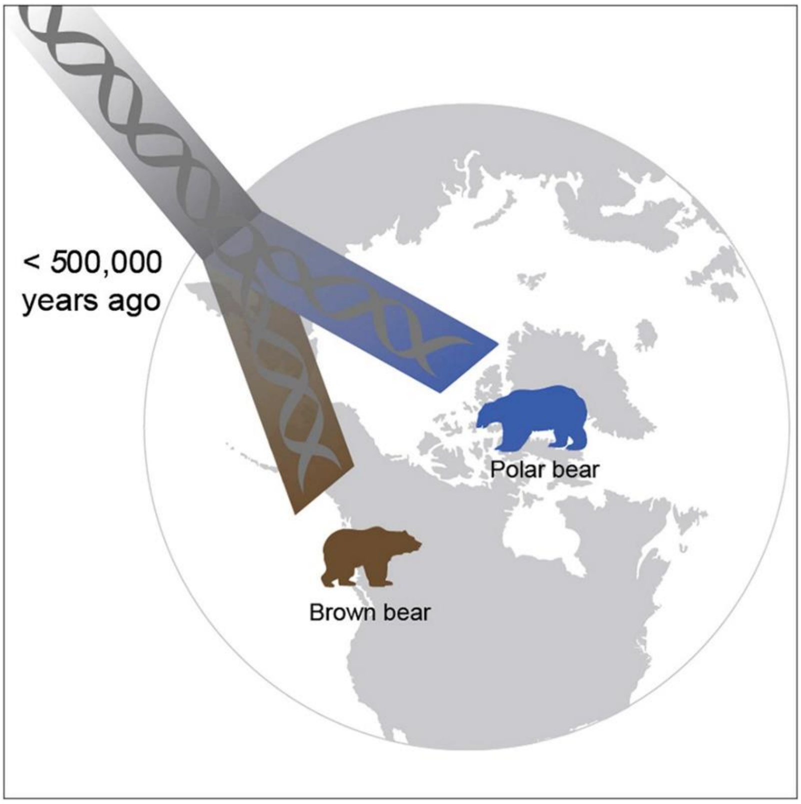


Fig. 9.1

The land masses that make up the polar bears habitat were once connected to North America, but over the past 60 million years these land masses have slowly shifted up in latitude till the Arctic Circle.

(a) Explain how polar bears diverged into a separate species from brown bears.

....

....

....

....

....

....

....

....[4]

1. Polar bears and brown bears have **a common ancestor** that inhabited land masses before they split up,

2. putting **ancestors under different selection pressures/natural selection acting differently on the two population/ Holoarctic exerted a different selection pressure than the Arctic regioins.**

3. **Gene flow & interbreeding** between bears in North America and the Arctic was **prevented due to geographical isolation/allopatric speciation.**

4. Differences in **allele frequencies accumulate through genetic drift and random mutations.**

5. Over time, the two populations accumulated enough phenotypic differences that they became **reproductively isolated/isolation of gene pool** such that they were not able **to interbreed to produce viable fertile offspring.**

Fossil of the bears ancestors' found in the Arctic caused scientists to initially estimated the divergence of these two species at 800 000 years ago. Recent mitochondrial genome sequencing data has suggested that the two species diverged less than 500 000 years ago, with polar bears exclusively colonising the Arctic Circle.

(b) Explain one advantage of using nucleotide sequences over morphology to establish phylogenetic relationships.

....

....

....

....[2]

1. Nucleotide sequences can be used to **compare organisms that look very similar** but are genetically different to **avoid the pitfall of convergent evolution./OWTTE**

2. Nucleotide sequences are **objective and quantifiable**, unlike morphological comparisons which are **subjective**.

3. Nucleotide sequence **differences can be used to predict relatedness accurately**, whereas morphology differences **cannot definitively predict degree of relatedness/divergence**.

4. Molecular methods, especially **mitochondrial genome comparisons**, can be used to establish **evolutionary history from the maternal lineage as it is very conserved**, but it is **impossible to use morphology to establish maternal lineage**.

AVP
Max 2

During the winter months bears hibernate as conditions in the environment become too extreme. Studies have shown that during hibernation, bear metabolism shows a preference for lipids as the energy source instead of carbohydrates.

(c) When triglycerides are hydrolysed, glycerol can be converted to a product that is an intermediate in glycolysis. Suggest what glycerol is converted into such that it can enter the glycolysis pathway.

....

....[1]

1. **Glyceraldehyde-3-phosphate**

(b) At certain times of the day oxygen levels dip, causing the hibernating bears to switch to anaerobic respiration for a short while.

Explain the significance of conversion of pyruvate into lactate during anaerobic respiration.

………

………

………

………

………

………[3]

1. In the absence of oxygen, **pyruvate acts as the alternative final electron acceptor**
2. And in the process of accepting the H+, **it regenerates NAD+ from NADH**
3. **NAD+ is required to allow glycolysis** to continue
4. To synthesize **nett 2 ATP via substrate level phosphorylation.**

[Total: 10]

10 Coral reefs are a vital source of income for fishermen. They provide fish fry with shelter and abundant food. They also protect coastlines from storms and being eroded. Yet each year, the range of coral cover is decreasing and its reason has been attributed to the enhanced greenhouse effect and climate change.

Fig.10.1 below shows the trend of sea surface temperature over the Great Barrier Reef from 1900 till 2020.

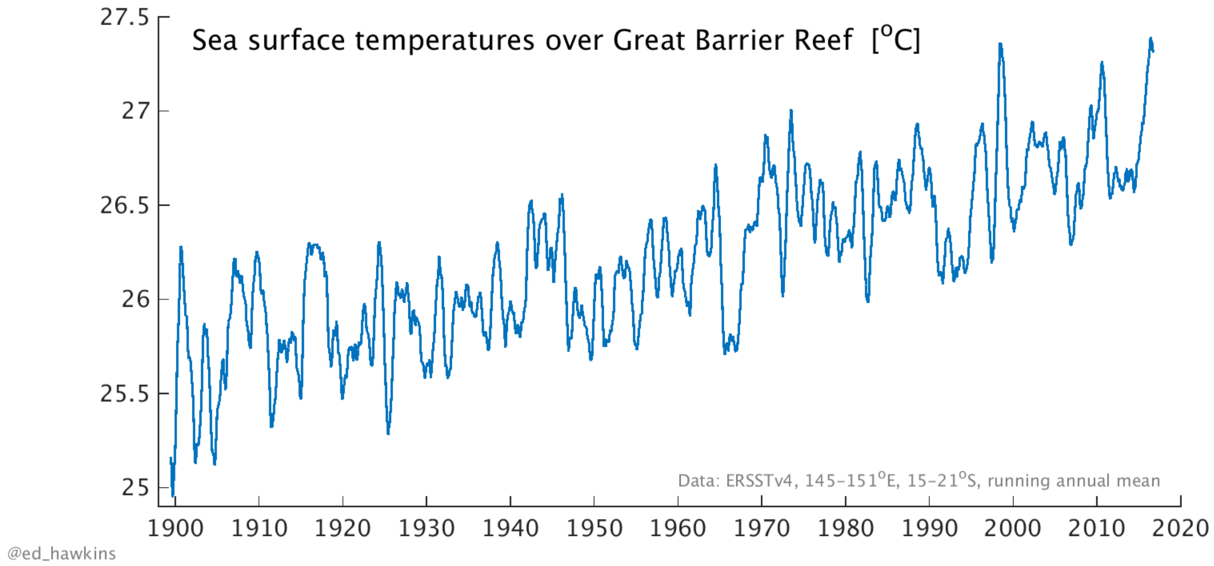


Fig.10.1

(a) Describe a human activity that has contributed to the enhanced greenhouse effect.

....

.... [1]

Any activity + greenhouse gas it produces.

1. **Burning of fossil fuels** for petroleum has led to **increased emissions of CO2**.
2. **Deforestation** of tropical forests removes trees that results in a **lowered rate of carbon fixation.**
3. **Rice Cultivation** flooding leads to soil emitting **methane.**
4. **Increased livestock rearing** leads to **increased methane production** from gas the animals' release.
5. **Fossil fuel extraction** process releases **large amounts of methane**.

AVP

(b) With reference to Fig.10.1, explain how the rise in ocean temperatures is linked to the reduction in coral reef cover.

....

....

....

....

....

....

....

....[4]

1. From **1900 to 2020**, there has been a **2.5 degree increase in the sea surface temperature**, from 25 degrees to 27.5 degrees.
2. This **increase in temperature stresses the zooxanthellae** living inside corals, who **produce toxins.**
3. The toxins stress the coral polyps, which **expel the zooxanthellae** and become **bleached/turn white**.
4. As the **temperature kept going up each year**, more of the **zooxanthellae do not return** to the corals which **have no source of food** now. More corals die each year, reducing the coral cover.

Plants found in the sea like kelp, seaweed and seagrass depend on photosynthesis for food production. In summer months where the days are longer, there is an increase in carbon dioxide concentration of oceans. Population of sea plants are expected to rise.

(c) Explain why the increase in sea plants is expected in the summer months.

....

....

....

....

....

.... [3]

1. **Increased carbon dioxide** in the oceans allows for an **increased rate of carbon fixation in** the Calvin cycle.

2. At the same time, because the **sun is out for a longer time**, light is **no longer a limiting factor** and **photophosphorylation occurs at a high rate**, producing ATP and NADPH.

3. More ATP and NADPH is available to match the rate of carbon reduction, producing a **higher quantity of GALP that can be used to produce more glucose required as a substrate for plant growth and metabolism.**

(d) Herbicides runoff from coastlines can cause sea plants to die and decrease in population. Many herbicides are photosystem inhibitors.

Explain briefly how herbicides cause the death of sea plants.

....

....

....

....[2]

1. When photosystems are inhibited, they can **no longer pass electrons** down the electron transport chain, **ATP and NADPH would not be produced**,

2. Both products that are **required for carbon reduction to produce GALP**, which is the raw material required for subsequent glucose production.

3. **No glucose is available for cellular respiration**, causing the plant to die as cellular activities cannot occur without ATP from respiration.

[Total: 10]

END OF PAPER